

ANKIT MEHTA

Atlanta, GA 30318 • (404) 955-3300 • amehta318@gatech.edu • amehta265.github.io • linkedin.com/in/ankit-h-mehta

EDUCATION

Georgia Institute of Technology

Aug 2026 – May 2028

M.S. Robotics, School of Interactive Computing: Concentration in Embodied AI & Mechanics

Georgia Institute of Technology

Aug 2019 – May 2023

B.S. Computer Science: Concentration in Artificial Intelligence & Information Networks | GPA: 3.79/4.00 | Highest Honors

RESEARCH

Robot Autonomy and Interactive Learning (RAIL) Lab | Dr. Sonia Chernova

Jul 2026 – Current

- Built a **VLM task planner** in ROS 2 for long-horizon household tasks on a Hello Robot Stretch 4 and a Fauna Robotics Sprout humanoid. It makes VLM queries at run time from the task, images, and memory, composes eight primitives (navigate, pick up, put down, observe, four analytic calls), and calls a skill after an object grounder confirms the target is in reach. [Link](#)
- Added **runtime failure detection and recovery**: monitor nodes at 10 Hz, 1 Hz, and 0.2 Hz check skill stop predicates and cancel a failed skill immediately. Memory keeps successful plans and the named failure modes with their recoveries. [Link](#)
- Integrated all the Stretch 4 sensor data across Zenoh and ROS 2 DDS (RTPS), and **cut image-processing latency by 180 ms**.

Borg Lab | Dr. Frank Dellaert

Jul 2026 – Current

- Implement a **hierarchical factor graph (STAG) in GTSAM** for autonomous mobile robots in a warehouse. Three levels carry learned latent states: fleet orchestration and task allocation, single-robot missions on the aisle graph, and motion and perception. Each level passes a sub-goal down to the level below and an encoded window of observations back up. [Link](#)
- Implement a **three-layer hierarchical control architecture** on continuous and discrete vector spaces for motion planning, trajectory planning, and feedback control. The motion layer and the feedback layer share one GTSAM factor graph. [Link](#)
- Built the warehouse simulator in **NVIDIA Isaac Sim**: pallet trucks and counterbalance AMRs drive through palletizing, staging, drop, marshalling, and outbound dock zones. The scene supplies the ground truth, wheel commands, and RTX sensor data.

Intelligent Vision & Automation Lab

May 2026 – Aug 2026

- Replaced a Kalman filter with a **GRU velocity predictor** for a non-holonomic robot in gap-based navigation. Trained the GRU on 922K samples with two new features, sector aspect ratio and density rate of change, that predict the gap congestion. [Link](#)
- The model **cut reactive planning failures by 30% across 350 episode runs**, and raised goal rate from 53% to 67% in a test of 50 episodes. Retuned three planner weights to cut mean path length from 52.7 m to 30.0 m. Rewrote two cost functions.

EXPERIENCE

Full Stack Software Engineer II | NCR Voyix

Jul 2023 – Jul 2026

- Built tree-based hierarchical caching systems for the role management user interfaces, and **cut onboarding time by 20%**.
- Led the software for a TypeScript payments integration for point-of-sale systems in APAC and EMEA, and built reusable, distributed Java uber-JARs so that the downstream client teams can use the integration in their own point-of-sale products.
- Built distributed services and edge computing software on GCP with Docker, Kubernetes, and Pub/Sub. Mentored engineers.
- Added facial recognition and ID pre-validation to retail self-checkout systems, and **cut average queue wait time by 23.6 s**.

Software Engineering Intern | Cypress.io

Aug 2022 – Dec 2022

- Built the Run All Specs feature (Vue, fuzzy sort) and In-App Cloud Integration (GraphQL, URQL, Docker). [Link 1](#), [Link 2](#), [Link 3](#)
- Built a nightly build system with GitHub Actions that tests each new Cypress binary against the pre-release packages. [Link](#)

PROJECTS

Swarm Robotics: Search & Rescue

Jun 2026 – Current

- Built a search-and-rescue system for two A1 quadrupeds in PyBullet with **no central controller or shared memory**. It uses PID control with gravity feedforward and a trot gait for locomotion, IR ray-cast perception, and a UDP protocol with covariance-intersection belief fusion for potential-field navigation and frontier-based area search in a simulated forest scene. [Link](#)

LLM Trading & Financial Chatbot

Sep 2022 – Mar 2026

- Compared LLM trading-agent performance on time-series data. Built an NLP portfolio chatbot for sector classification, stock recommendations, and market charts. The portfolio strategy **won first prize in the 2023 BlackRock Challenge**. [Link 1](#), [Link 2](#)

PUBLICATIONS

Mehta, A. et al. (2022). *Hate Speech and Offensive Language Detection in Ethiopian Languages*. International Conference on Learning Representations (ICLR) Workshop. Research done as part of the TID Lab at Georgia Tech. arxiv.org/abs/2208.09941

TID Lab (2021 – 2023): built a Naïve Bayes language classifier in fastText for low-resource Ethiopian languages, applied graphSAGE to classify inflammatory users, and used BERT for post similarity in Aggie, an open-source election platform. [Link](#)

SKILLS

Languages & Tools: Python, C++, Java, TypeScript, JavaScript, Golang, C, React, Vue, Docker, Kubernetes, GCP, GraphQL, Git

Robotics & ML: ROS 2, GTSAM, Isaac Sim, MuJoCo, PyBullet, PyTorch, TensorFlow, scikit-learn, OpenCV, SAM2, Hugging Face

Concepts: Embodied AI, Task Planning, Factor Graphs, State Estimation, Navigation, Imitation & Reinforcement Learning, NLP